

Unit 1, Lesson 2: Ratio Relationships

Benchmarks

MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b , or $a:b$, where $b \neq 0$.

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

Learning Targets

- ☐ I can interpret ratios when given multiple representations.
- ☐ I can solve real-world problems involving ratios.

1.2.1 Warm-Up: Equivalent Fractions

1. Using the digits 1 to 9 once each, fill the boxes so that the three equations show equivalent fractions.

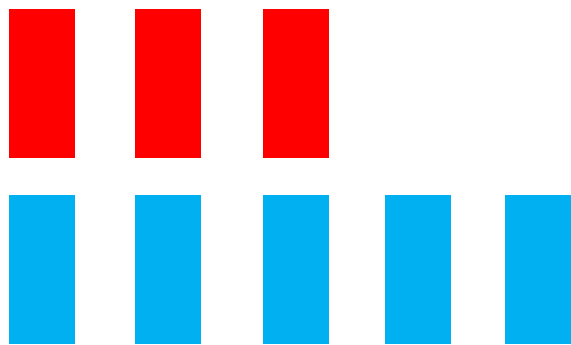
$$\frac{7}{\square 2} = \frac{\square 8}{\square}$$

$$\frac{3}{\square} = \frac{\square}{10}$$

$$\frac{12}{\square 6} = \frac{2}{6} = \frac{\square}{2 \square}$$

1.2.2 Refresher: Reading and Writing Ratios

1. The Florida Gators are experimenting with creating a new flavor of Gatorade, Gatormax. To create Gatormax, they will mix the following ratio of red and blue Gatorade together.



- a. Complete the statement.
- There are ____ red Gatorades for every ____ blue Gatorades in Gatormax.

There are three ways to express ratios.

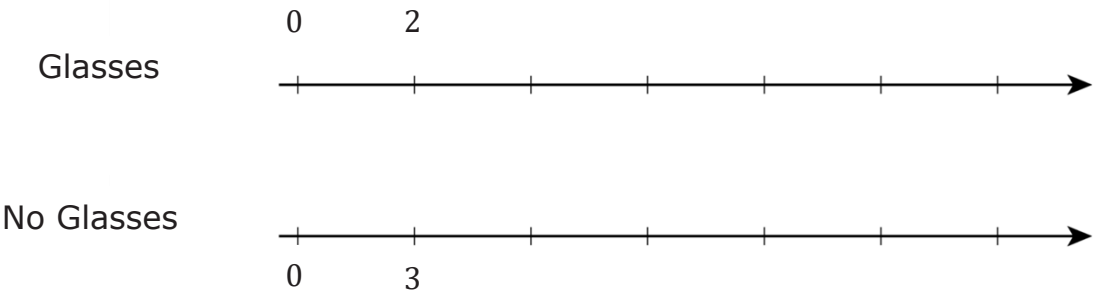
Method for Writing Ratios	The Ratio of Red Gatorades to Blue Gatorades
Using a colon (:)	3 : 5
Using the word "to"	3 to 5
Using a fraction bar	$\frac{3}{5}$

- b. Write the ratio of blue Gatorades to total Gatorades in three ways.

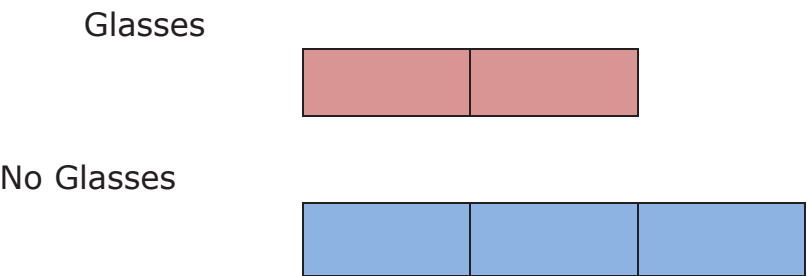
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1.2.3 Guided Practice: Representing Ratio Relationships

1. The ratio of the number of students who wear glasses to the number of students who do not is 2:3.
- a. Complete the double number line for this ratio.



- b. If 18 students in the class do not wear glasses, how many students are in the class?
- c. Use the tape diagram to determine how many students do not wear glasses given that there are 25 students in the class.



- d. Complete the table.

Glasses	2		
No Glasses	3	24	
Total Students			80

1.2.4 Collaboration: Rates and Unit Rates

1. Four students work as baristas after school. They all start out with the same pay per hour. During the first year of employment, the baristas earn a \$0.05 per hour raise in pay each month. Work with your partner to complete the following questions.
- a. Use the relationship between the number of hours worked and the amount of money earned to complete the table.

Employee	Money Earned (\$)	Hours Worked	Rate (\$ per hours)	Unit Rate (\$ per hour)
Elijah	62	4		=
Joy	39	2.5		=
Sharon	173.25	11		=

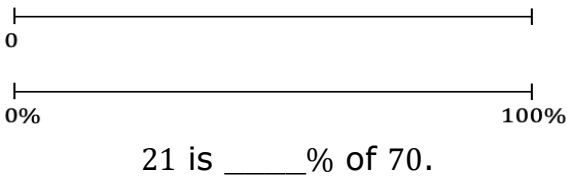
- b. Which student has worked as a barista for the longest amount of time?
- c. Brian was hired a month after Elijah. What is Brian’s rate of pay per hour?
- d. Brian worked 6 hours last week. How much money did Brian earn last week?

1.2.5 Guided Practice: Applying Ratios to Percentages

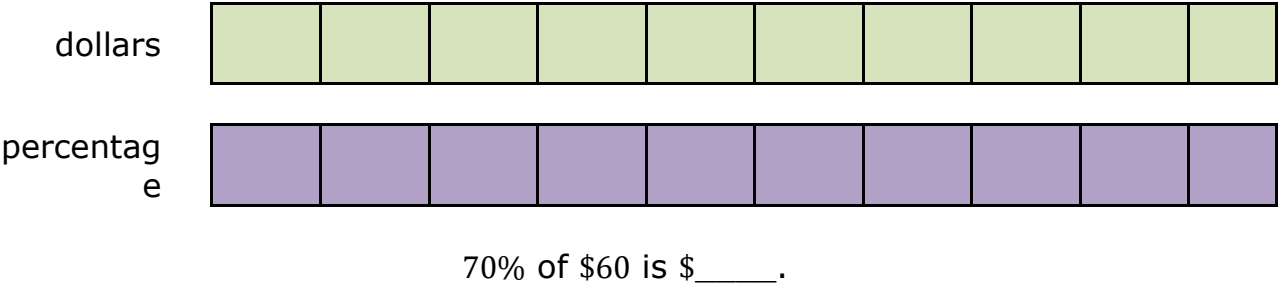
A **percentage** is a ratio expressed as a fraction of 100.

Work with your partner to use the diagrams or tables provided to solve the problems.

1. Determine what percentage of 70 is 21. Then, complete the statement.



2. Determine 70% of \$60. Then, complete the statement.



3. A pair of jeans is on sale for 60% of its regular price. The regular price of the jeans is \$65.
- a. Complete the ratio table to determine the sale price.

Price (\$)	Percentage
	10
	60
	100

- b. Complete the statement.
- The sale price of the jeans is \$_____.

1.2.6 Wrap-Up: Ticket Out the Door

1. The table shows the cost of boxes of crayons at a convenience store.
- a. Use the ratio table to find the cost of 15 boxes of crayons.

Crayon Boxes	6		
Cost (\$)	9		

- b. Find the unit rate in dollars per box.
2. Fatima has a 20% off coupon at the store. Her total comes out to \$70. Use a double number line, tape diagram, or ratio table to determine how much her coupon is worth.

